

AI-Powered Document Intelligence

Project proposal — Retrieval, embeddings, and intelligent document analysis

1. Problem Statement

Organizations often store large collections of reports, research papers, and technical documents that are difficult to search manually. A system is needed to retrieve relevant information and provide concise, grounded answers.

2. Proposed Solution

The proposed platform ingests documents, extracts text, creates embeddings, stores vectors in a searchable index, retrieves relevant passages, and uses a language model to generate answers with source context.

3. Objectives

Build a reliable document ingestion pipeline; implement semantic retrieval; support question answering over uploaded documents; expose source passages; and evaluate answer relevance and retrieval quality.

4. Technology Stack

Python, FastAPI, an embedding model, a vector database or FAISS, a large language model, and a web-based interface. The architecture can be extended with OCR for scanned documents.

5. Expected Outcome

The final system should provide fast semantic search and grounded responses while reducing the time required to locate useful information across large document collections.

6. Future Scope

Future versions may include agentic workflows, document comparison, automatic citation generation, multimodal document understanding, feedback-driven retrieval, and enterprise access control.

Technical reference document — generated sample